The universe as a thing: a trilemma for Brading's P1**

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Abstract

Brading's local present is a spatiotemporal region of whatever size is needed to sustain a dynamical system. Her P1** requires every existing thing's present to be less than the whole history, and she says that "in order to avoid the block universe, all we need is (a)." This note grants her identification of what exists with the genuine unities and her claim that the law-constitutive approach applies to the material universe. It argues that the universe case is underexplored. If the universe is a genuine unity and is in P1**'s domain, its sustaining region is either the whole history, a proper region (spatially global or local), or outside the domain. On the first, (a) is false of the universe; on the second, Brading's rejection of a spatially global present needs qualification, or the local region must sustain the universe's unity without the relations among all its members; on the third, P1**'s quantifier is restricted and the global claim is lost. The result is a trilemma, not a refutation: P1** cannot be both universal and satisfied by the universe without an account of the universe's sustaining region that Brading does not supply. The 2013 defeat condition and the 2015 survival condition treat subsystem existence as the decisive empirical question, but the universe case is independent of it. A four-event model also shows that, even for proper subsystems, (a) and (b) do not imply a proper union. The verdict is limited: (a) does not establish Brading's sufficiency sentence, and the missing premise is an account of how the universe's own present is fixed.

1 The texts and the question

Katherine Brading's *Presentism as an Empirical Hypothesis* and *Physically Locating the Present* offer one of the most careful attempts to keep a presentist ontology alive within special relativity (Brading, 2013, 2015). She rejects the assumption that spacetime is the ontological principle of unity of what exists, and replaces it with a law-constitutive principle: the dynamical laws constitute genuine unities, and the present of a system is the spatiotemporal region needed to sustain its dynamics. The 2013 paper introduces the idea in a compact form:

> The present is a spatiotemporal region of whatever size is necessary to sustain the dynamical system in question (Brading, 2013, p. 1109).

The size is left open. It "might turn out to be much, much smaller than the abstract spatiotemporal structure within which the evolution of that unity is completely described, or it might be the size of the block universe" (Brading, 2013, p. 1109). Brading then assumes genuine subsystems and draws the local conclusion:

> So let us begin from the position that there are genuine subsystems. This means that the size of the spatiotemporal region required to sustain the system is less than the

> entire block universe, and so the present is (at least in the first instance) local, not global (Brading, 2013, p. 1110).

The 2015 paper states the position as P1**:

> (P1**) For each and every thing that exists, that thing exists only presently, where for each thing the spatiotemporal extent of the present for that thing is dependent on the dynamics and is (a) less than the entire spatiotemporal evolution of the universe as represented in Minkowski spacetime, and (b) less than the entire spatiotemporal evolution of that thing as represented in Minkowski spacetime (Brading, 2015, p. 17).

She adds: "In order to avoid the block universe, all we need is (a)" (Brading, 2015, p. 17). Her own statement of the effect is weaker: a smaller-than-universe present "calls the ontology of a block universe into question" (Brading, 2015, p. 17). She also leaves the empirical question open: whether there are genuine subsystems "smaller in spatiotemporal extent than the entirety of spacetime" is a matter of the details of the dynamics, and "only if there are such can the version of presentism offered here survive" (Brading, 2015, p. 18).

This note asks what (a) establishes for the universe itself. The answer is a trilemma, not a refutation. Granting Brading's identification of what exists with the genuine unities, the universe is a candidate genuine unity. If it is in P1**'s domain, its present must be less than the whole history; but the region needed to sustain it may be the whole history, a proper region (spatially global or local), or outside the domain. Each option has a cost. If the universe is excluded, the quantifier is restricted and the global claim is lost. The sufficiency sentence "all we need is (a)" does not settle which option holds.

The note does not claim that the coverage problem is wholly new, and it does not compete with Savitt's 2021 objection. It claims that the universe case is a distinct problem, that it is generated by Brading's own texts, and that her survival condition does not address it. The formal model in §5 is a second, independent problem for the subsystem family.

2 What exists are the genuine unities

Brading's law-constitutive account is not only an account of the unity of ordinary objects. She says that the principle applies across the board:

> We are being offered a principle of ontological unity that applies across the board to simple things, composite systems, and component parts of those systems, and which applies all the way up to the material universe as a whole. Insofar as what exists is what is one, the law-constitutive approach to unity is a metaphysical thesis about what exists (Brading, 2015, p. 16).

The "insofar as" is a genuine conditional. Brading does not argue for the law-constitutive approach here; she refers the reader to earlier work (Brading, 2015, p. 16). The unconditional formulation appears on the next page: "we begin from the premise that what exists (what is real) are the genuine unities," and "what is 'present' depends on what is real (i.e. what exists, the genuine unities), and what is real depends on the dynamical laws" (Brading, 2015, p. 17). The identification is a working premise of her argument, not a theorem. This note therefore grants it. The question is what follows for the universe if the premise is granted, not whether the premise is true.

Granting the premise, two claims are needed for the universe to be a thing that exists. The first is that the universe is a genuine unity. Brading's scope claim—the principle applies "all the way up to the material universe as a whole"—gives the principle's intended scope, not the verdict that the universe satisfies it. A principle of unity can apply to a collection and deliver the verdict that the collection is not one. The textual support for the verdict is elsewhere:

> The dynamics grounds the unity of what there is, both of the parts and of the whole (consisting of interacting parts) (Brading, 2013, p. 1110).

and the 2013 footnote says the law-constitutive approach covers "the unity of simple things, composite things, and the totality of existing things," applying to all three (Brading, 2013, p. 1105, n. 3). The same footnote adds that "different principles of unity might be used for the different cases," so the scope claim is not a uniform criterion. It is a licence to treat the whole as a unity under some principle.

The second claim is that the universe satisfies the relevant principle. Brading associates unity with the conservation of total quantity of motion when a system is isolated from external interactions (Brading, 2013, p. 1105), and with a composite whole that conserves its total quantity and direction of motion "when that system as a whole is free from external interactions" (Brading, 2015, p. 16). The universe is the limiting case of an isolated system: there is no external interaction by definition. This is an argument, not a quotation, and a reader may resist the application of the isolation condition to the totality. The note treats the universe's unity as an interpretive premise that Brading's texts support but do not compel. If the premise is rejected, the universe falls outside the domain, which is the third horn of the trilemma.

3 The universe's present: a trilemma

Grant the two premises: the universe is a genuine unity and a thing that exists. P1** is universal: "for each and every thing that exists" (Brading, 2015, p. 17). What is the spatiotemporal extent of the universe's present? Brading's proposal is that the present is the region "of whatever size is needed to sustain the dynamical system in question" (Brading, 2015, p. 17). The region needed to sustain the universe is not fixed by the text. Three options are available if the universe has a present, and each has a cost. Brading uses "local" for any present that is a proper part of the block (Brading, 2015, p. 17). For the trichotomy below I use "local" more narrowly, for a proper region that is not spatially global; the narrower sense is the one at issue.

A prior possibility should be noted. Brading's present is the region needed to sustain a *dynamical system*. If the universe is a genuine unity but not a dynamical system in that sense—no system/environment distinction, no external interactions, no parameter of evolution—it may have no sustaining region at all. Then P1** either fails for it (it exists but has no present) or is undefined for it. That is a cost of the same kind as the horns below; the horns concern the case in which the universe has a present. I return to it in §8.

(1) The present is the whole history $\mathcal{W}$. If the region needed to sustain the universe is the entire spatiotemporal evolution, then the universe's present is $\mathcal{W}$. Clause (a) requires every existing thing's present to be less than $\mathcal{W}$. The universe is an existing thing. So (a) is false of the universe, and P1** is false on its unrestricted reading. This is the strongest horn for the critic. Brading herself says that if a system's needed region "is the entire spatiotemporal history of the universe," then "we are back to the block universe" (Brading, 2015, p. 17). The universe is a candidate system for which that description is apt.

(2) The present is a proper region. A proper region avoids the contradiction with (a). It divides into two sub-cases.

(2a) Spatially global. A closed system's evolution is determined by its state on a Cauchy slice, and the slice is a proper region of the history. If "needed to sustain" means "the data needed to determine the evolution," the universe's present could be a slice or a thick global slab, not $\mathcal{W}$. Its cost is Brading's own rejection of a spatially global present. She writes that "the idea of a spatially global present common to all systems is not preserved in this version of presentism, but the rejection of a spatially global present is one of the central lessons of special relativity" (Brading, 2015, p. 17). The qualifier "common to all systems" leaves room for a global present for the universe alone, while subsystem presents are local and non-transitive. But the second clause rejects a spatially global present without that qualifier. At a minimum, Brading owes an account of why the universe's global slice is not the global present she rejects. The text does not give one.

(2b) Local. A local patch might sustain the universe's unity if the laws determine the rest from the patch. Its cost is that the patch must sustain the unity of a whole whose members are spread through the history. On Brading's account of aggregation, "the members must stand in relations to one another, and those relations must be necessary and sufficient for the collection to form a unity" (Brading, 2015, p. 16). If the relations are spatiotemporal and span the history, a local patch omits them. If the relations are dynamical and encoded in the patch, the patch can sustain the unity, but then the account owes an explanation of how a local region encodes the relations among all members. Brading does not give one. The sub-case is not excluded; it is underexplored. Other proper non-global regions—light cones, disconnected unions, slabs with holes—are local-data candidates if their data plus the laws determine the rest; if they do not, they are not sustaining regions. The patch is the representative case.

(3) The universe is not in the domain. This horn divides.

(3a) The universe is not a genuine unity. Under the granted identification of what exists with the genuine unities, the universe is then not an existing thing. P1** does not apply because there is no thing to quantify over. The cost is not the global claim; it is that the universe does not exist, which contradicts Brading's scope claim that the principle applies "all the way up to the material universe as a whole" (Brading, 2015, p. 16).

(3b) The universe is a unity but "thing" is a term of art. If "thing" is restricted to proper unities below the whole, the universe is a unity but not a "thing" in P1**'s sense. The quantifier is restricted, and the cost is the global claim: P1** says "for each and every thing that exists," and the universe is left out. The view then says that every proper subsystem exists only presently, not that the whole of reality does. That is a different thesis.

Among the cases in which the universe has a present, the three horns are exhaustive: either it is in the domain and its present is $\mathcal{W}$; or it is in the domain and its present is a proper region, either spatially global or local; or it is not in the domain. If it has no present, P1** fails or is undefined for it, as noted above. No case yields P1** as stated without a further account. Horn (1) makes (a) false of the universe. Horn (2a) requires a qualification of Brading's rejection of the global present. Horn (2b) requires an account of how a local region sustains the whole. Horn (3a) denies the universe's existence; horn (3b) gives up the global claim.

The conclusion is therefore conditional, not categorical. P1** cannot be both universal and satisfied by the universe without an account of the universe's sustaining region that Brading does not supply. The sufficiency sentence "all we need is (a)" does not tell us which horn holds. The note does not claim that no such account can be given; it claims that Brading does not give one, and that the two obvious proper-region candidates each have a cost.

The trilemma is stated in the common representation $\mathcal{W}$, as the coverage model is. Brading's 2013 note 7 denies determinate ontological spatial or temporal relations between systems: "All the ontology is carried by the dynamics, and we frame the dynamics spatiotemporally, but ontologically, the dynamics requires no such spatiotemporal underpinning" (Brading, 2013, p. 1109, n. 7). If that denial applies to the universe's present as well, the trilemma is a result about the representation, not about the ontology. The dialectic is internal to Brading's text, which uses the Minkowski representation; a reader who rejects the common frame should read the trilemma as a conditional about that frame.

4 The defeat condition and the survival condition

Brading's 2013 defeat condition describes a different scenario:

> It might turn out that, for whatever system we try to consider, the size of the spatiotemporal region necessary to sustain it is the entire history of the universe, encompassing all that ever has been and ever will be. In other words, the only dynamical system that there is, is this entirety, and there are no genuine subsystems of the universe. If that turns out to be the case, then this version of presentism is defeated and the block universe triumphs (Brading, 2013, p. 1109).

"For whatever system we try to consider" is a universal quantifier. If every system's region is $\mathcal{W}$, then no system has a proper region, so there are no proper subsystems. The "in other words" is valid in that direction. It is not an equivalence: the absence of proper subsystems does not entail that the only system's region is $\mathcal{W}$; the only system might have a proper region.

The scenario in §3 is different. The universe's region may be $\mathcal{W}$ while subsystems have proper regions. The subsystems are parts of the universe, not alternatives to it. Brading's defeat condition addresses the universal scenario; it does not address the universe-own-region scenario. That is not a logical defect in a sufficient condition. It is a gap in the framing.

The 2015 text makes the framing explicit:

> Even were this approach to succeed, the crucial question would remain of whether there are genuine subsystems of the world, as picked out by the dynamical laws, that are smaller in spatiotemporal extent than the entirety of spacetime. Only if there are such can the version of presentism offered here survive (Brading, 2015, p. 18).

And 2013: "A great deal rests on whether there are genuine subsystems of the universe, and that is something that we can find out only through empirical inquiry" (Brading, 2013, p. 1110). Brading treats the existence of subsystems as the decisive empirical question. The universe case is independent of it. Even if there are genuine subsystems, the universe remains a candidate genuine unity with a present to fix. The survival condition is necessary for the local presents of subsystems; it is not the question on which the truth of P1** turns.

Three scenarios should be kept apart:

1. *Universal:* every system's region is $\mathcal{W}$; there are no proper subsystems; Brading's stated sufficient condition for defeat is met.

2. *Universe-own-region:* the universe's region is $\mathcal{W}$ while subsystems have proper regions; the defeat condition is not met, but (a) is false of the universe if it is in the domain.

3. *Subsystem-only:* the universe is excluded or not a unity; subsystems have proper regions; P1** is restricted and the global claim is given up.

Brading's empirical question decides whether (1) holds and whether (3) is available. It does not decide (2), which is the case that generates the trilemma.

5 The retreat to subsystems and the coverage model

The natural reply to the trilemma is horn (3b): restrict P1** to proper subsystems. The retreat has one cost: it gives up the global claim. P1** says "for each and every thing that exists." If the universe is excluded, the quantifier is restricted, and the view no longer says that every existing thing exists only presently. It says that every proper subsystem does. That is a different thesis, not a reading of the original.

Independently, a defender who keeps the global claim and tries to secure it through (a) and (b) for subsystems faces a second problem: even for proper subsystems, (a) and (b) do not imply a proper union. Clause (a) says, for every system i,

$$\mathcal{U}_i \subsetneq \mathcal{W}.$$

It does not imply

$$\bigcup_i \mathcal{U}_i \subsetneq \mathcal{W}.$$

Proposition 1 (Non-implication)**.** *Local boundedness of every $\mathcal{U}_i$ is consistent with $\bigcup_i \mathcal{U}_i = \mathcal{W}$.*

Proof. Take $\mathcal{W} = \{a, b, c\}$ and the proper subsets $\{a, b\}$ and $\{b, c\}$. Their union is $\mathcal{W}$. □

The natural reply is that clause (b) closes the gap. Clause (b) says that each thing's present is less than the entire spatiotemporal evolution of that thing. That is a bound on the present relative to the thing's own history. It is not a bound on the present relative to every other thing's present, and it is not a claim that some event lies outside every present.

Proposition 2 ((a) and (b) do not imply a proper union)**.** *There is a model in which (a) and (b) hold for every system and $\bigcup_i \mathcal{U}_i = \mathcal{W}$.*

Proof. Let $\mathcal{W} = \{a, b, c, d\}$. Let system 1 have history $\mathcal{H}_1 = \{a, b, c\}$ and present $\mathcal{U}_1 = \{a, b\}$. Let system 2 have history $\mathcal{H}_2 = \{b, c, d\}$ and present $\mathcal{U}_2 = \{c, d\}$. Then $\mathcal{U}_1 \subsetneq \mathcal{W}$ and $\mathcal{U}_2 \subsetneq \mathcal{W}$, so (a) holds. Also $\mathcal{U}_1 \subsetneq \mathcal{H}_1$ and $\mathcal{U}_2 \subsetneq \mathcal{H}_2$, so (b) holds. But $\mathcal{U}_1 \cup \mathcal{U}_2 = \{a, b, c, d\} = \mathcal{W}$. □

Three assumptions must be stated. First, the model is set-theoretic, not dynamical: it does not show that $\mathcal{U}_i$ is the region *necessary to sustain* system i. It models the extensional shadow of (a) and (b). Second, "less than" is an extent comparison, and $\subsetneq$ is a regimenting choice; it assumes containment, which is a charitable strengthening. Third, the model uses a common event domain $\mathcal{W}$. Brading's 2013 note 7 denies determinate ontological spatial or temporal relations between systems: "All the ontology is carried by the dynamics, and we frame the dynamics spatiotemporally, but ontologically, the dynamics requires no such spatiotemporal underpinning" (Brading, 2013, p. 1109, n. 7). The model therefore treats $\mathcal{W}$ as the common epistemic representation in which the comparison is stated, not as a claim that the systems stand in determinate ontological relations. If no common representation is allowed, the model is a result about the representation, not about the ontology.

With those assumptions, the model shows that per-item properness is coverage-neutral: (b) puts events outside their *own* system's present, not outside *every* present. A defender who wants non-coverage needs an event $e \in \mathcal{W}$ such that for all i, $e \notin \mathcal{U}_i$. Neither (a) nor (b) delivers one. Non-coverage requires a joint constraint on the family, or a selection principle that removes some presents. The model is a burden-shift, not a refutation of Brading's dynamics.

The relation between the two problems is this. The trilemma concerns the universe; the coverage model concerns the subsystem family. A defender who restricts the quantifier to avoid the trilemma gives up the global claim. A defender who keeps the global claim must answer the universe case, and the coverage model shows that the subsystem clauses do not by themselves secure the global restriction. The two problems are distinct, and neither is a substitute for the other.

6 Scale-general, not scale-free

The diagnosis is not that Brading lacks an existence criterion. She has one: what exists are the genuine unities (Brading, 2015, p. 17). Nor is the criterion literally scale-free. Her 2013 note says "different principles of unity might be used for the different cases," while the law-constitutive approach "applies to all three" (Brading, 2013, p. 1105, n. 3). The approach is *scale-general*: it is offered as applicable to simple things, composite systems, and the totality, with different principles possible at different scales.

Scale-generality explains why the trilemma is not resolved by the criterion alone. The approach does not guarantee that its dynamical selection procedure picks out a proper scale. Brading says genuine subsystems are "as picked out by the dynamical laws" (Brading, 2015, p. 18), and she associates unity with conservation of total quantity of motion when a system is isolated (Brading, 2013, p. 1105). That is a selection procedure. What is missing is a guarantee that the procedure selects a proper subsystem rather than the maximal unity. The 2013 defeat condition is exactly the case in which it selects no proper subsystem. The universe-own-region case is the case in which it selects the maximal unity *and* proper subsystems; the text does not say which present the maximal unity has.

The missing premise is therefore not a criterion of existence but a scale-selection guarantee, or an account of the maximal unity's present. A presentist who wants the global restriction must add a privileged actuality, a co-presence relation, or a relative existence predicate. Each is a further structure with a cost; none is P1**.

Relative existence is developed by Balaguer's R-presentism and Mason's domain- and level-dependent empirical presentism (Balaguer, 2026; Mason, 2026), and Rovelli's "neither presentism nor eternalism" defends a local, unoriented becoming (Rovelli, 2019). These are coherent positions, but they change the question: every event exists for some system or frame, and the global claim is given up. They are not restorations of P1**.

7 Relation to Savitt, Saunders and Pooley

Savitt's 2021 objection is distinct and remains unanswered. He argues that Brading's local present is "a characterization of a relativistic present, not relativistic presentism," because every dynamical system has a sustaining region and "it is unclear under what conditions two dynamical systems share a present" (Savitt, 2021, §3.3). That is a population and co-presence objection: many presents, no public present. It is not the universe trilemma. Savitt does not discuss Brading

2015, P1**, or the law-constitutive principle's application to the universe. The two arguments are compatible: Savitt gives the semantic verdict, and the trilemma gives the metaphysical pressure internal to Brading's system.

Saunders's requirement that the presentist provide an invariant, definable equivalence relation of co-presence is the sharpest version of the co-presence cost (Saunders, 2000); the definability results make a non-trivial symmetric equivalence relation of a present hard to come by (Clifton and Hogarth, 1995). It supports Savitt's objection, not the trilemma. Pooley's distinction between a deflationary analysis of passage compatible with the block and a privileged global present that does not take relativity seriously (Pooley, 2013) is about passage, not about the domain of existence. Brading's paper is about existence, so Pooley's dilemma does not apply directly. The trilemma does not claim that local presents fail to account for passage.

The interaction passage is sometimes read as a system-relative existence claim. Brading writes: "we belong to the same world as whatever we interact with, and the rest of that world is as real as we are," and asks, "Does my son exist relative to me when he is in London and I am here? Of course" (Brading, 2013, p. 1110). The passage is better read as a single-domain claim: "the rest of that world is as real as we are" is not indexed to a system, and the following question is about the *determinacy* of what the son is doing, not about his existence. It does not supply a relativized quantifier, and it does not answer the trilemma.

8 Objections and limits

The identification is a premise, not a theorem. The note grants Brading's identification of what exists with the genuine unities. It does not argue that the identification is true, and it does not need to: the question is what follows for the universe if the identification is granted. A reader who rejects the identification is outside the argument. A reader who accepts it must still decide whether the universe satisfies the unity condition; that is the interpretive premise discussed in §2.

The universe's present may be a proper global slice. This is the strongest reply to horn (1). A Cauchy slice is a proper region and determines the evolution of a closed system. If Brading accepts a global slice for the universe alone, horn (2) holds, and the trilemma's cost is a qualification of her rejection of the spatially global present, not a contradiction. The note does not claim that horn (1) is forced. It claims that the text does not settle which horn holds, and that each horn has a cost.

The local-patch horn. A local patch whose laws determine the rest would make (a) true of the universe. The note does not claim that no such account can be given. It claims that Brading does not give one, and that the aggregation condition makes the account non-trivial: the patch must sustain the unity of a whole whose members are spread through the history.

The no-present case. If Brading's present requires a dynamical system with an environment, the universe may not qualify, and then it has no sustaining region. P1** would be false or undefined for it. This is a cost of the same kind as the horns, not a fourth region. The note does not decide whether the universe is a dynamical system; it flags the assumption that the trilemma's horns share.

Priority. The note does not claim that the universe trilemma is wholly new, and it does not claim that no one has noticed the tension. It claims that the trilemma is generated by Brading's texts, that it is distinct from Savitt's objection, and that it has not been answered. A published statement of the same trilemma would defeat the priority claim, not the argument.

What the note does not show. It does not show that presentism is false, that eternalism is true, or that Brading's physics is wrong. It does not show that the universe's present is $\mathcal{W}$, or that there are no genuine subsystems. It shows that P1** is false of the universe on the whole-history horn, true only with an unspecified proper-present account on the proper-region horns, or restricted on the quantifier horn; and that the subsystem clauses do not secure a proper union.

9 Conclusion

Brading's P1** is her strongest anti-block formulation. It is also, by her own principle, under pressure from the universe. What exists are the genuine unities; the approach applies all the way up to the material universe; the universe is a candidate genuine unity and a thing that exists. Its present is either the whole history, a proper region (spatially global or local), or outside the domain. The first makes (a) false of the universe; the global sub-case needs a qualification of Brading's rejection of the global present; the local sub-case needs an account of how a local region sustains the whole; the third gives up the global claim. The trilemma is not a refutation, but it is a standing problem for the sufficiency sentence.

The 2013 defeat condition and the 2015 survival condition treat subsystem existence as the decisive empirical question. The universe case is independent of it. Even if there are genuine subsystems, the universe remains a candidate genuine unity with a present to fix. The empirical question decides between the universal and subsystem-only scenarios; it does not decide the universe-own-region scenario.

The retreat to subsystems gives up the global claim, and even for subsystems (a) and (b) do not imply a proper union. The missing premise is a scale-selection guarantee, or an account of the maximal unity's present, not another bound on the extent of presents. A presentist who wants the global restriction must add a privileged actuality, a co-presence relation, or a relative existence predicate. Each is a further structure with a cost; none is P1**.

References

Mark Balaguer. *How to Be a Presentist*. Oxford University Press, 2026. doi: 10.1093/9780197845745.001.0001. Chapter: How to Make Presentism Consistent with Special Relativity.

Katherine Brading. Presentism as an empirical hypothesis. *Philosophy of Science*, 80(5):1101–1111, 2013. doi: 10.1086/673897.

Katherine Brading. Physically locating the present: A case of reading physics as a contribution to philosophy. *Studies in History and Philosophy of Science Part A*, 50:13–19, 2015. doi: 10.1016/j.shpsa.2014.09.007.

Rob Clifton and Mark Hogarth. The definability of objective becoming in Minkowski spacetime. *Synthese*, 103(3):355–387, 1995. doi: 10.1007/BF01089733.

Lucy Mason. Temporal perspectives, probabilities, and openness. *Synthese*, 2026. doi: 10.1007/s11229-026-05493-1. Preprint: PhilSci-Archive 25374 (2025).

Oliver Pooley. Relativity, the open future, and the passage of time. *Proceedings of the Aristotelian Society*, 113(3):321–363, 2013. doi: 10.1111/j.1467-9264.2013.00357.x. Preprint: PhilSci-Archive 9936.

Carlo Rovelli. Neither presentism nor eternalism. *Foundations of Physics*, 49(12):1325–1335, 2019. doi: 10.1007/s10701-019-00312-9. arXiv:1910.02474.

Simon Saunders. Tense and indeterminateness. *Philosophy of Science*, 67(S3):S600–S611, 2000. doi: 10.1086/392849. Preprint: PhilSci-Archive 2241.

Steven F. Savitt. Being and becoming in modern physics. In Edward N. Zalta, editor, *The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University, 2021. URL https://plato.stanford.edu/entries/spacetime-bebecome/. Substantive revision 6 October 2021; §3.3 "Localizing Time".